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RA-IVX en regresiones casi integradas: asimetría y aplicación a la curva de Phillips

Trabajo de Fin de Grado

Enrique Ruiz de Almirón Da Silva

NIU: 100476072

Tutor: Nazarii Shalis

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Predictive regressions are among the most widely used tools in economics and finance. Their goal is to determine whether a variable observed in the present contains information about the future evolution of another variable. While these regressions have long been part of the standard toolkit of economic forecasting, the specific set of econometric problems addressed in this thesis has a much more concrete origin, one that emerged from within financial economics.

Following the work of Fama and French (1988) and Campbell and Shiller (1988), an extensive literature took up the question of whether future asset returns could be predicted using valuation variables observed in the present, such as dividend yields or price-earnings ratios. The underlying hypothesis was simple: relatively low valuations should be associated with higher future returns. However, the very nature of the predictors used to answer this question would eventually give rise to a substantial econometric literature devoted to studying the properties of these regressions and the validity of the inference drawn from them.

The reason is that these financial predictors shared two problematic features. On one hand, their values moved with extreme slowness. On the other, their innovations were strongly correlated with those of returns themselves. Under this combination, the OLS slope estimator inherits a finite-sample bias, and the distribution of its t-statistic departs from the standard normal, so that tests of predictability ended up over-rejecting Stambaugh (1999). The literature has responded along two paths that ultimately converge: bias correction methods and the recovery of standard normal inference. Within the second path, the instrumental variables method IVX of Kostakis, Magdalinos and Stamatogiannis (2015) has become the reference, and building on this method, Demetrescu and Rodrigues (2022) construct the RA-IVX estimator, which combines both corrections by augmenting the regression with the predictor's residuals. There is one aspect of this bias that the literature has not focused on, simply because it was not its object of study. The Stambaugh bias has a sign: under the typical negative correlation of financial predictors, it tends to be positive, and since in those applications the expected effect is also positive, the bias operates in the same direction as the signal one is trying to detect. The opposite region, where the expected effect is negative, has received less attention.

In this thesis we examine the problem of predictive regressions with highly persistent regressors and the various solutions offered by the literature. We review, through a Monte

Carlo simulation, the finite-sample behavior of RA-IVX against IVX and OLS, focusing on the asymmetric advantage that arises when the true effect is negative. Finally, we take the method outside its natural habitat: to the Phillips curve, where theory predicts a negative relationship between unemployment and inflation.

1. Econometric Challenges in Predictive Regressions

Predictive regressions with persistent regressors generate two related but logically separate problems. The first is a finite-sample problem; that is, the OLS estimator of β is systematically biased. The second is asymptotic in nature and persists regardless of the amount of data available: even after correcting the bias, the t -statistics used to assess significance fail to behave well. The predictive regression we study, together with the process governing its predictor, are given by:

$$r_{t+1} = \alpha + \beta x_t + u_{t+1}, \quad (1)$$

$$x_{t+1} = \mu + \rho x_t + v_{t+1}, \quad (2)$$

where r_{t+1} denotes future excess returns, x_t the predictor observed at t , and ρ the parameter governing its persistence. Depending on this parameter, we could have stationarity, a unit root, or a whole range of intermediate possibilities. Two features of this system can generate distortions. The first is the high persistence of financial ratios, a slow dissipation of shocks that makes them hard to distinguish from a random walk. The second is that the innovations in the equations above, u_{t+1} and v_{t+1} , are mechanically correlated, since both depend on asset prices (the predictor is built from prices, and returns are determined by price changes). We will see in the following sections that neither feature is fatal on its own; it is only together that they generate the inferential difficulties at the heart of this literature.

1.1. Finite-Sample Bias of the OLS Estimator

Under these features, Stambaugh (1999) showed that the OLS estimator of β carries a finite-sample bias. This is not a standard endogeneity problem, since x_t and u_{t+1} are orthogonal

across time, and OLS therefore remains asymptotically consistent. The failure is subtler. In a standard i.i.d. regression, orthogonality between the regressor and the error term is enough to guarantee unbiasedness. Each observation is an independent draw, so the sample mean inherits the population moment and OLS behaves as it should. In time-series econometrics, however, observations are no longer independent draws. Depending on the degree of persistence, each x_t can depend on the previous observation, which in turn depends on the one before it, and so on. The denominator of the coefficient, $\sum x_t^2$, therefore ceases to be a fixed object and becomes one accumulated along the same realized path as the numerator, $\sum x_t u_{t+1}$.

What determines how much the numerator and denominator co-move depends on two conditions. The first is $\text{corr}(u, v) \neq 0$; that is, the return innovation and the predictor innovation must be partly driven by the same shock, so that v_{t+1} shows up both in the numerator through u_{t+1} and in the denominator through x_t^2 itself. The second is $\rho \approx 1$, meaning the predictor must be persistent enough for v_{t+1} to propagate forward and remain embedded in $x_{t+2}, x_{t+3}, \dots$, the path of x_t . If either condition fails, the covariance will tend to zero and orthogonality is again sufficient.

Projecting u_{t+1} onto v_{t+1} and substituting into the OLS ratio yields Stambaugh's (1999) derivation, which splits the bias into two terms. One captures the part of the return innovation that is orthogonal to the predictor; it leaves no trace in the path of x_t and therefore vanishes in expectation. The other is a more problematic component: a ratio of correlated random quantities, $\sum x_t v_{t+1} / \sum x_t^2$, which is inherited from the OLS estimation error of ρ in the AR(1), $\hat{\rho} - \rho$. Its expectation does not vanish even though the numerator has mean zero. This expectation is quantified by Kendall (1954) as $\mathbb{E}[\hat{\rho} - \rho] \approx -(1 + 3\rho)/T$ near the unit-root neighborhood. Further details are given in Appendix A. Stambaugh (1999) ultimately shows that the bias formula can be written as:

$$\text{Bias}(\hat{\beta}) = \frac{\sigma_{uv}}{\sigma_v^2} \cdot \mathbb{E}[\hat{\rho} - \rho] \quad (3)$$

For valuation ratios, the sign is unambiguous. The Kendall bias is negative (it underestimates ρ), and σ_{uv} is also negative, since a rise in asset prices increases returns while lowering the price ratio. The product of the two is positive, so OLS overstates predictability. The implication of this derivation is that the bias has a known analytical form that can be estimated and subtrac-

ted from $\hat{\beta}$. This is the logic followed by bias-correction methods. We will see, however, that correcting the point estimator is not enough for inference. That problem has a different origin, as the next section explains.

1.2. Non-Standard Asymptotics: Why the t -Test Fails

Suppose we correct the point estimator by subtracting the Stambaugh bias; the inference problem will still remain, since its nature is different.¹ Even with a centered estimator, the t -statistic used for hypothesis testing does not converge to a standard normal distribution when working with persistent predictors, and we therefore cannot rely on Gaussian critical values.

Where does this problem come from? For most financial predictors, it is not possible to know empirically whether ρ is strictly less than or equal to one. Unit-root tests have low power against local alternatives when ρ lies in the local-to-unity region. Elliott, Rothenberg and Stock (1996) showed that even the most powerful unit-root tests have a limited ability to detect stationarity in these regions, particularly with the limited number of observations we actually work with in practice. We thus find ourselves stuck in a gray zone where no asymptotic framework truly applies, and forcing a choice risks misspecifying the model.

The local-to-unity framework takes a different approach: rather than assuming ρ is known, it treats the degree of persistence itself as the object of uncertainty. Phillips (1987) introduced it, and Cavanagh, Elliott and Stock (1995) brought it into predictive regressions. The idea is to write the autoregressive coefficient as $\rho_T = 1 + c/T$ with $c \leq 0$, so that the root approaches one as the sample grows, with the rate governed by c : we get an exact unit root when $c = 0$, and increasing stationarity as c becomes more negative.

Under this parametrization, the partial sum of x_t converges to a process known as an Ornstein–Uhlenbeck process, J_c . One could think of it as a Brownian motion, but pulled back toward its mean with a force proportional to how negative c is. When $c = 0$, there is no mean-reverting force, and the process behaves as a true Brownian motion. Likewise, the more negative

1. The bias, after all, is only a finite-sample problem. Why? Because under fixed stationarity, the Kendall bias goes to zero as $T \rightarrow \infty$, $\mathbb{E}[\hat{\rho} - \rho] \approx -(1 + 3\rho)/T \rightarrow 0$, so the contamination disappears with enough data.

c becomes, the stronger the mean-reverting force (i.e., the stationarity). Since J_c depends on c , different values of c produce different processes and, in turn, different limiting distributions for any statistic built from x_t . This is precisely what ends up contaminating the t -statistic. Following Cavanagh, Elliott and Stock (1995), the limiting distribution of t is:

$$t_{\hat{\beta}} \Rightarrow \delta \cdot \frac{\int_0^1 J_c(r) dW_v(r)}{\left(\int_0^1 J_c(r)^2 dr\right)^{1/2}} + \sqrt{1 - \delta^2} \cdot Z, \quad (4)$$

where $\delta = \sigma_{uv}/(\sigma_u \sigma_v)$ is the correlation between the shocks, $Z \sim N(0, 1)$, and J_c is the process governed by c . The second term is standard normal and harmless, but the first is not, since it contains J_c , and because c cannot be consistently estimated, we do not know the distribution of the t -statistic or which critical values should be used. Two cases are worth noting. If $\delta = 0$, the non-standard term disappears and we can rely on standard Gaussian critical values; this is the only case in which OLS inference is safe. In the other case, $\delta \neq 0$, which covers any relevant valuation ratio, the distribution will be non-standard and conditional on the unknown c . The challenge, then, is to carry out valid inference without knowing c . There are two main answers: Bonferroni approaches like the Campbell and Yogo's (2006) one, and the IVX framework developed by Kostakis, Magdalinos and Stamatogiannis (2015).

1.3. Bias-Correction Methods

The natural starting point is to exploit the fact that the bias has a known analytical form. Since the bias is given by $\text{Bias}(\hat{\beta}) = (\sigma_{uv}/\sigma_v^2) \cdot \mathbb{E}[\hat{\rho} - \rho]$, we could estimate these components from the data and subtract the bias from $\hat{\beta}_{OLS}$. This correction was already present in Stambaugh (1999), included alongside the original result.

Amihud and Hurvich (2004) tackle the problem in the same spirit but from a different angle. Their idea is that if v_t were observable, it could be included directly as a regressor: $y_t = \alpha + \beta x_{t-1} + \gamma v_t + \varepsilon_t$. In this augmented regression, the new error term ε_t would be orthogonal to x_{t-1} once the endogenous part has been included as a control, and $\hat{\beta}$ would therefore be exactly unbiased. However, v_t is not observable, so the equation above is infeasible in practice. The solution proposed by AH is to replace the term with bias-corrected residuals $\hat{v}_t^*$, obtained using

a corrected autoregressive coefficient $\hat{\rho}^* = \hat{\rho} + (1 + 3\hat{\rho})/T + \dots$. This correction is essential, since the ordinary OLS residuals carry the Kendall bias into $\hat{v}$. As long as $\hat{\rho}^*$ is sufficiently accurate, the t -statistic of $\hat{\beta}$ in the feasible augmented regression converges to $N(0, 1)$ both under stationarity and near-integration. The main limitation is that this accuracy deteriorates near the unit root, exactly where the correction matters most. The guarantee is therefore conditional: it only holds if we can estimate persistence well. Some of the methods discussed in the next sections will not need this at all.

Lewellen (2004) took a different route that avoids estimating ρ altogether. Estimating ρ reliably is hardest precisely in the cases where the correction matters most, so he chose to sidestep this circularity. How? By being conservative. He imposes $\rho = 1$ directly in Kendall's approximation, which gives $\mathbb{E}[\hat{\rho} - \rho] \approx -4/T$; that is, Lewellen considers the worst-case scenario. If the true $\rho < 1$, this produces a small overcorrection, but we gain robustness against misspecifying persistence. Lewellen showed that this conservative adjustment would substantially weaken much of the prior evidence on return predictability.

These three methods, each in its own way, correct the point estimator. The inference problem, however, remains when ρ is close to one, since, as we saw, the distribution of the t -statistic depends on an unknown c . The next section considers another procedure that takes the uncertainty about c as its starting point.

1.4. Robust Inference Procedures: Campbell and Yogo (2006)

The methods above tackle the problem from the estimation side, improving $\hat{\beta}$. Campbell and Yogo (2006) start from a different place, focusing on inference instead. Before describing the CY construction, it is worth ruling out the most natural intermediate answer: estimating c from the data, inverting an ADF test, and conditioning inference on that point estimate. This does not solve the problem, however, since the imprecision of $\hat{c}$ shifts the critical values of c .² Campbell and Yogo (2006) avoid estimating c altogether through a Bonferroni construction.

² Cavanagh, Elliott and Stock (1995) formalize the following argument: if the limiting distribution of our t -statistic is sensitive to c at exactly the same scale at which $\hat{c}$ remains imprecise, then using $\hat{c}$ in place of c shifts the critical values by an amount that never becomes negligible.

The first step is to bound c rather than estimate it: inverting a unit-root test over a grid of values yields a confidence interval (c_l, c_u) that contains the true c with probability $1 - \eta_1$. The second step exploits the fact that, conditional on c , the limiting distribution of t is fully characterized by equation (4). It depends on c and δ , but on no other unknown quantity. So for each c within (c_l, c_u) , Campbell and Yogo invert a bias-corrected t -statistic to obtain a confidence interval for β ; they then take the union over all values of c , so that the Bonferroni inequality guarantees:

$$P(\beta \notin CI_\beta) \leq \eta_1 + \eta_2 = \eta,$$

This means the overall test controls size at the desired level η without requiring a point estimate of c . But this robustness is not free; power is affected. The total significance level η must be split between the two stages: η_1 goes toward bounding c , and η_2 toward testing β . Whatever significance is allocated to the first step can no longer be used to detect predictability. Uncertainty about c does not invalidate the test, but it does reduce its ability to detect predictability.

There is, however, a more fundamental limitation in Campbell and Yogo. Valid inference within this framework requires the researcher to commit to a particular class of persistence for the predictor. In the end, the localization expression is a specific modeling assumption, and the test is only valid within that assumption. If the true process approaches unity at a rate other than $1/T$, the local-to-unity asymptotics no longer apply. This motivates an inference procedure that remains valid regardless of the predictor's persistence and of the specific rate at which it approaches unity. That is the strategy proposed by Kostakis, Magdalinos and Stamatogiannis (2015).

1.5. The IVX Framework

As discussed, CY's main limitation is not simply the loss of power from bounding c , but also that the researcher cannot remain agnostic about the true process driving the predictor. This is because the framework allows for different degrees of persistence through c , while fixing the rate at which the coefficient approaches one, which raises a natural question: could valid inference on β be carried out without even taking a stance on persistence? Kostakis, Magdalinos and Stamatogiannis (2015), building on the theoretical work of Magdalinos and Phillips (2009), answer with an elegant idea: rather than trying to pin down how persistent the predictor is

and adjusting for it, they construct an instrument whose distribution is asymptotically normal regardless of the true process governing the predictor. They begin by building the instrument from the first difference of the predictor. Since x_t follows $x_t = \rho x_{t-1} + v_t$ with $\rho \approx 1$, the first differences $\Delta x_t = x_t - x_{t-1} \approx v_t$ are approximately stationary, thereby removing the local-to-unity component. These increments are then fed into an artificial autoregression in which the researcher chooses an artificial root R_{nz} . The following expressions characterize the method:

$$\tilde{z}_t = R_{nz} \tilde{z}_{t-1} + \Delta x_t, \quad \tilde{z}_0 = 0, \quad R_{nz} = 1 - \frac{c_z}{n^\delta}, \quad c_z > 0, \quad 0 < \delta < 1. \quad (5)$$

The key idea is best seen by comparing the two roots: while the predictor has root $\rho_T = 1 - c/T$, the instrument uses $R_{nz} = 1 - c_z/T^\delta$. Since $T^{-\delta}$ decays more slowly than T^{-1} when $\delta < 1$, the term being subtracted is larger for any finite T , so the instrument's root stays farther from unity than the predictor's. The instrument still converges toward unity (which makes it relevant), but more slowly than the regressor. It is precisely this intermediate persistence regime that Magdalinos and Phillips (2009) termed mild integration, and it is the property that makes IVX work. The intuition is analogous to the cancellation of an unknown scale in a pivotal statistic.³

Why does OLS fail? We said that the persistent predictor x_{t-1} is correlated with the regression error u_t . This happened because the same shock driving x_t , namely v_t , was correlated with u_t , and due to high persistence this correlation accumulates indefinitely. As a result, x_{t-1} and u_t become asymptotically entangled in the t -statistic, in the sense that the numerator and denominator of the t -test cannot be separated, and the distribution depends on the unknown c . IVX breaks this by replacing x_{t-1} with the instrument $\tilde{z}_{t-1}$, artificially constructed to forget

3. The reason IVX is able to restore standard inference is best seen through an analogy. When estimating a normal mean with unknown variance σ^2 , we divide by $\hat{\sigma}/\sqrt{T}$ to obtain a pivotal statistic. We do not actually know the variance, but the denominator contains exactly the same one as the numerator, so it cancels in the ratio. IVX follows the same logic, except that the unknown object that cancels is not a scalar but the problematic stochastic process J_c encoding the predictor's unknown persistence. To be precise, the instrument does not eliminate J_c from the estimation; we still have a dependence on c . What the instrument guarantees is that the process enters the numerator and denominator of the Wald statistic with an identical structure, so that it can cancel in the ratio, something OLS cannot achieve.

its past quickly enough that its correlation with u_t vanishes asymptotically. The instrument is built from $\Delta x_t \approx v_t$, so it remains correlated with x_{t-1} , which guarantees relevance, but, due to the mild integration property, its memory decays fast enough that, in the limit, its partial sums converge to a Brownian motion independent of u_t . This independence means the uncertainty of the IVX estimator can be written as a single unknown scale, estimated through an object $\hat{Q}_H$. The Wald statistic is then divided by $\hat{Q}_H$, canceling the scale, and what remains has a χ^2 distribution that does not depend on c . This robustness, however, does not come without cost. Replacing x_{t-1} , which accumulates signal over T periods, with the mildly integrated $\tilde{z}_{t-1}$, which accumulates signal over T^{δ_z} , means working with less information. The test loses power relative to what could be achieved if we knew c . The parameter δ_z controls this trade-off: values closer to one preserve more power but also let some persistence leak through, while values closer to zero maximize robustness at a substantial cost to power. In practice, they recommend $\delta_z = 0,95$.

Despite the advantages of IVX, another finite-sample limitation persists. Hosseinkouchack and Demetrescu (2021) show that the statistic $\tilde{t}_{ivx}$ carries a residual non-centrality in finite samples. While the derivation is involved, one could say it stems from the fact that constructing the instrument z_{t-1} from filtered Δx_t does not fully remove the existing correlation between the regressor's innovations (v_t) and those of returns (u_t) in finite samples, so that the instrument, despite being designed to control for persistence (mild integration), still inherits part of that contemporaneous correlation that gives rise to the classical endogeneity bias, the same underlying mechanism as in Stambaugh (1999).

Demetrescu and Rodrigues (2022) take up this argument and offer an observation: if there is a systematic shift in $\tilde{t}_{ivx}$, and since $\hat{\beta}_{IVX} - \beta = \tilde{t}_{ivx} \cdot se(\hat{\beta}_{IVX})$ and the standard error is, by construction, always positive, this translates into a bias in $\hat{\beta}_{IVX}$, which motivates their proposal, combining the bias-correction intuition of Amihud and Hurvich (2004) with the instrumental-variables framework of KMS.

The starting point, as in AH, is the infeasible augmented regression:

$$y_t = \alpha + \beta x_{t-1} + \gamma v_t + \varepsilon_t, \quad (6)$$

By including v_t , we orthogonalize ε_t , which makes $\hat{\beta}$ exactly unbiased. The feasible procedure

replaces v_t with the simple OLS residuals $\hat{v}_t$ obtained from an $AR(p)$ for x_t . We proceed in two stages: first, we regress $\ddot{y}_t$ on $\hat{v}_t$ by OLS, $\tilde{y}_t = \ddot{y}_t - \hat{\gamma}\hat{v}_t$. We then apply IVX to regress $\tilde{y}_t$ on $\ddot{x}_{t-1}$, obtaining $\tilde{\beta}_{IVX}$. The main difference from Amihud and Hurvich (2004) is that, instead of using bias-corrected residuals $\hat{v}_t^*$, we use simple OLS residuals. In the AH procedure, residual quality is crucial because OLS is sensitive to how precisely $\hat{\rho}$ is estimated. In Demetrescu and Rodrigues (2022), the IVX instrument breaks this dependence. The instrument converges more slowly than OLS, so the error in $\hat{v}_t$ shrinks relative to the estimation signal. The same slow convergence that lowers IVX's power is what makes the augmentation robust to the quality of the residuals.

Demetrescu and Rodrigues improve inference along two channels. First, since the non-centrality depends on the correlation between u_t and v_t , replacing u_t with the orthogonalized residual $\varepsilon_t = u_t - \gamma v_t$ eliminates (infeasible case) or attenuates (feasible case) the source of the correlation, better centering the distribution of $\tilde{t}_{ivx}$ and improving size. This in turn reduces the bias of $\hat{\beta}_{IVX}$. Second, by projecting the v_t component out of the regression error, the residual variance falls from σ_u^2 to $\sigma_\varepsilon^2 < \sigma_u^2$, raising the signal-to-noise ratio and making genuine predictability more likely to be detected. This power gain is not guaranteed in every case, however, since $se(\tilde{\beta}_{ivx})$ picks up an additional term, $\hat{\gamma}^2 \hat{Q}_T$, which captures the cost in uncertainty of not being able to work with the true v_t and having to estimate it instead.

2. Monte Carlo Evidence

Demetrescu and Rodrigues do not focus on the negative range of betas, since in their case of interest (predictive regressions in finance) they expect positive theoretical betas. There is reason to think, however, that RA-IVX could be especially advantageous in the case of negative betas. Hosseinkouchack and Demetrescu show that, under a negative correlation between u_t and v_t , the distribution of the t -statistic carries a non-centrality that shifts it to the right. By identity, this would translate into a systematically positive bias of $\hat{\beta}_{IVX}$, regardless of the sign of the true beta, and correcting this bias could generate an asymmetric advantage in power. We examine, through a Monte Carlo experiment, the finite-sample behavior of OLS, IVX and RA-IVX under conditions reflecting the empirical settings found in the literature, investigating this potential asymmetry. The data are generated from: $y_t = \beta x_{t-1} + u_t$ with $x_t = \rho x_{t-1} + v_t$, where $(u_t, v_t)' \sim N(0, \Sigma)$, with

unit variances and contemporaneous correlation $\rho_{uv} = -0,9$. Persistence is fixed at $\rho = 0,99$, placing x_t firmly in the near-integrated regime. The instrument follows the KMS specification with $c_z = 1$ and $\delta_z = 0,95$. Results are based on 2,000 replications for the power analysis and 5,000 for the distributional analysis. Our sample sizes span $n \in \{50, 100, 150, 200, 300, 500\}$.

Figure 1 shows the empirical rejection rates under $\beta = 0$ for different values of persistence and sample size. The pattern confirms the theoretical predictions from Section 2: OLS over-rejects severely at $\rho = 0,99$, exceeding 20% at $n = 50$. IVX delivers a substantial improvement, staying between 6–8% even in the local-to-unity region, though it remains above the 95% simulation band. RA-IVX appears to be the only procedure that stays within the band, with rejection rates between 3.5–5.5% across all values of ρ considered, confirming that augmenting with $\hat{v}_t$ removes the residual non-centrality that IVX retains in small samples. We can also see that as n grows, the size distortion of OLS coming from the Kendall-Stambaugh bias shrinks, since the Kendall bias tends to zero, $\mathbb{E}[\hat{\rho} - \rho] \approx -(1 + 3\rho)/T \rightarrow 0$.

Figure 2 shows the empirical distributions of the t -statistic, always constructed to test the null hypothesis but under different data-generating processes (three different values of β) for $n = 100$. Under $\beta = 0$, we can see the non-centrality parameter directly in the mean: OLS = 1,24, IVX = 0,79, and RA-IVX = 0,38. Here OLS performs worst, followed by IVX, due to the non-centrality that RA-IVX manages to partially correct. This explains the over-rejection documented in Figure 1.

As discussed, this positive non-centrality also translates into a positive bias of $\hat{\beta}_{IVX}$, and we expect this bias to shrink under the augmented regression. Table 2 reports the mean bias and the mean standard error across the three experiments. We find that the biases are of roughly the same magnitude and sign in all three cases, and that all of them shrink under the augmentation. We also observe an improvement coming from variance reduction.

What we therefore expect is that the effect on power is asymmetric depending on whether we face a theoretically positive or negative β . Under $H_0 : \beta = 0$, IVX's positive bias translates into over-sizing, and under positive alternatives $\beta > 0$, that same bias adds to the true drift, generating spurious excess rejection (above the correct theoretical power). Under negative alternatives $\beta < 0$, we expect the opposite: the positive bias, when added to the true (negative)

drift, pulls the t -statistic toward the critical value, reducing the probability of rejection. The test would therefore be losing power to detect genuine predictability. This effect of non-centrality on bias is asymmetric, unlike the effect of variance.

The variance effect, by contrast, is symmetric, in the sense that a reduction in the standard error increases the absolute value of the t -statistic and therefore helps capture both a positive and a negative drift.⁴ We therefore expect the following pattern in the t -distributions:

In the positive case, we find two offsetting effects: on one hand, the variance reduction lets us better capture a positive beta, but on the other, the attenuation of the positive bias reduces the t -statistic. These opposing effects offset one another, so we expect the means of the t -statistics not to move much. On the negative side, by contrast, both effects pull in the same direction. The variance reduction better captures the negative drift, and the removal of the positive bias pushes beta further negative. We therefore expect the RA-IVX t -statistic to move further away from the IVX t -statistic.

What we expect theoretically is confirmed by the means of the distributions shown in Figure 3. In the positive case, the mean of t_{ivx} exceeds that of t_{ra} by only 0.12, a modest difference, as anticipated. Under the negative beta, by contrast, the mean of t_{ra} is 0.75 more negative than that of t_{ivx} , which is already a sizeable magnitude. This confirms the channel we expected: by retaining the positive bias, IVX under-rejects the negative alternative, failing to detect as significant negative signals that genuinely are.

The power curves (Figure 2) confirm the anticipated asymmetry. RA-IVX clearly dominates IVX in power under negative alternatives, while the difference under positive alternatives favors IVX due to weaker size control. In Table 3 we observe that power at $n = 50$ is nearly double for alternatives at or below -0.075 . For instance, at -0.1 we are able to detect the alternative 21 % of the time with RA-IVX versus 11 % with IVX. This advantage fades as n grows and t_{ivx} approaches its asymptotic normal distribution, which we can see in the convergence of the curves as n increases.

4. The term comes from stochastic process theory. We call drift the systematic shift in the mean of the limiting distribution of the t -statistic attributable to the true value of beta under a local alternative, distinguishing it from a shift of a different origin (a finite-sample artifact, not dependent on beta).

Overall, the Monte Carlo evidence confirms the theoretical motivation: the non-centrality of $\tilde{t}_{ivx}$ under a fixed negative correlation generates a positive bias in $\hat{\beta}_{IVX}$ regardless of the sign of the true beta. Eliminating this bias through augmentation consistently corrects test size under the null across all three cases, and works to restore power for negative betas while reducing the spurious power of $\hat{\beta}_{IVX}$ for positive signals. It is interesting how this interacts with the variance-reduction effect, which raises power symmetrically. This power gain on the positive side is partly offset by the reduction in positive bias; on the negative side, by contrast, both the variance reduction and the reduction in beta's bias work together, which is where we expect the largest finite-sample advantages.

3. An Empirical Application: The Phillips Curve

The econometric methodologies discussed in Section 1 originated within financial economics. However, the same pathologies can arise in any predictive regression with highly persistent regressors. The Phillips curve is a natural example. The unemployment rate is structurally near-integrated, with autoregressive roots close to unity over standard macroeconomic sample horizons.

What makes the Phillips curve especially useful as an empirical application of predictive regressions is how the finite-sample bias operates. In the financial-ratios literature, we saw that a negative correlation between innovations generates a positive bias once multiplied by the (also negative) Kendall bias. The same mechanism operates in the Phillips curve: since unemployment and inflation tend to move in opposite directions, this generates a negative correlation and, therefore, a positive Stambaugh bias. Here, however, a key idea emerges: economic theory predicts a negative β ; excess supply in the labor market lowers wages and price pressures, so the same positive bias that implies an overstatement of predictability in the classical cases could have the opposite effect in the Phillips curve, attenuating the negative coefficient toward zero.

The direct consequence is that OLS can push estimates toward zero, failing to detect predictability not because the true relationship is absent, but because the bias suppresses the signal below conventional significance thresholds. The IVX estimator, while correcting OLS's inconsistency under persistence, is not exempt from an analogous problem in finite samples. The

positive non-centrality of the t_{ivx} statistic under negative correlation, documented by Hosseinkouchack and Demetrescu (2021), carries over into a positive bias of $\hat{\beta}_{IVX}$ itself, which in a sense mimics the behavior of the Stambaugh (1999) bias by reducing the significance of negative betas. Against this backdrop, RA-IVX, by attenuating the source of the non-centrality (the effective correlation of the errors), recovers higher power than both OLS and IVX.

This empirical exercise becomes a natural test: in the subsamples where diagnostics predict attenuation, will RA-IVX recover significance that neither OLS nor IVX attain?

3.1. Model, Data, and Estimation Strategy

The predictive regression takes the following form: $\pi_{t+1} = \alpha + \beta x_t + \varepsilon_{t+1}$

United States. Inflation is constructed as $\pi_t = 100 \cdot \Delta \log(\text{CPI}_t)$, where CPI_t is the Consumer Price Index for All Urban Consumers (series CPIAUCSL). The unemployment rate x_t is taken directly in levels (series UNRATE). Both series are drawn from the FRED-MD dataset of McCracken and Ng (2016).

Euro Area. Inflation is constructed as $\pi_t = 100 \cdot \Delta \log(\text{HICP}_t)$, where HICP_t is the Harmonised Index of Consumer Prices. The unemployment rate x_t is the total Euro Area unemployment rate. Both series are drawn from the EA-MD-QD dataset of Barigozzi, Lissona and Tonni (2024). No transformations beyond the log difference are applied to inflation.

The diagnostics in Table 4 confirm that the regression falls within the setting of predictive regressions with highly persistent regressors. The asymmetry between the two variables is clear. Inflation is stationary in most of the samples: the ADF test rejects the unit root at conventional significance levels, with AR(1) coefficients ranging between 0.32 and 0.64. The unemployment rate is not: its AR(1) coefficients exceed 0.94 in every sample, reaching as high as 1.003 in the Euro Area, and the ADF test fails to reject the unit root in every sample except the full U.S. monthly sample.

Not all subsamples, however, are equally valid for inference. The pre-Volcker U.S. subsample should be set aside if we want to be rigorous. Inflation in that period fails the ADF test, with a p -value of 0.248 and AR(1) coefficients of 0.648. The IVX framework requires a stationary dependent variable, and if inflation itself is near-integrated, the predictive regression

no longer fits the asymptotic theory for which IVX was developed.

The estimated correlation between innovations, $\widehat{\text{corr}}(u, v)$, are modest in absolute magnitude, implying that finite-sample biases are present but not large. We expect RA-IVX to recover significance relative to OLS and IVX in the subsamples where endogeneity is more pronounced, and to coincide with them where it is not.

We apply three estimators to each regression: OLS, IVX, and RA-IVX. OLS is estimated by least squares on the demeaned series, with homoskedastic standard errors, and is included as a biased benchmark. IVX follows the calibration of Kostakis, Magdalinos and Stamatogiannis (2015): a mildly integrated instrument is built from the predictor, and the point estimator $\hat{\beta}_{\text{IVX}} = \sum_t z_{t-1} \tilde{\pi}_{t+1} / \sum_t z_{t-1} \ddot{x}_t$ is obtained, whose inference is robust to the persistence and endogeneity of the regressor. The recursive construction of the instrument and the standard-error formula are given in Appendix B. RA-IVX (Demetrescu and Rodrigues, 2022) proceeds in two steps prior to IVX. First, we fit an $\text{AR}(p)$ to the predictor by OLS and store its residuals $\hat{v}_t$, with p selected by AIC. Second, we augment the dependent variable $\hat{v}_t$ and apply IVX to the augmented regression, obtaining $\tilde{\beta}_{\text{IVX}}$ with heteroskedasticity-robust variance. The full derivation is given in Appendix B.

3.2. Results

3.2.1. Euro Area

The Euro Area results tell a coherent story. The full sample yields robust and significant predictability for all three estimators, with coefficients clustered around $\hat{\beta} = -0,042$. The sign is consistent with theory: higher unemployment predicts lower inflation. Moreover, the agreement across methods is itself informative: with $\widehat{\text{corr}}(u, v)$ close to zero, the bias-correction machinery has little to correct, and all three estimators behave exactly as theory predicts when endogeneity is negligible.

The subsample results should be read with caution. The pre-2008 period spans 93 months versus 202 for the post-2008 period, which considerably limits statistical power. Coefficients are somewhat smaller in absolute value ($-0,036$ versus $-0,042$), but the lack of significance in the earlier subsample cannot be attributed to a structural absence of predictability. What the results

do firmly establish is that Phillips-curve predictability is statistically robust in the post-crisis period for all three methods.

In the European case, the low endogeneity and the small change in the betas, consistent with low correlations, suggest that the instrument barely corrects anything. Nevertheless, the efficiency cost of instrumental variables discussed in Section 1.5 (a product of slower convergence) is paid in full, as we can see in the increase in standard errors. This illustrates the robustness cost that is paid even when, ex post, there turns out to be little endogeneity to correct. It is precisely in other subsamples, where that endogeneity is more relevant, that this cost finds its counterpart.

3.2.2. United States

The U.S. results present a more nuanced picture than the Euro Area, with substantial heterogeneity across subperiods that the full sample conceals. No significant predictability is found in the full sample, with t -statistics well below conventional thresholds across all three methods and a $\widehat{\text{corr}}$ of $-0,041$. This is not surprising, since averaging across regimes with opposing dynamics mechanically dilutes any subsample-specific signal.

The Volcker–Greenspan subsample delivers the most striking results, with positive and significant coefficients across all three estimators. This sign is counterintuitive from the perspective of a demand-driven Phillips curve, but it is consistent with the empirical literature on the Volcker period. The Volcker period was dominated by supply shocks that raised unemployment and inflation simultaneously, breaking the standard negative relationship. The three estimators do not differ much from one another. In the subsequent periods we enter the case we document, with negative betas and negative correlation between innovations.

The Great Moderation produces no significant results under any method, consistent with the widely documented flattening of the Phillips curve during this period (Blanchard et al., 2015). Stable inflation and well-anchored expectations left insufficient variation for unemployment to have detectable predictive power. The Phillips curve did not structurally disappear; it became statistically invisible because monetary policy successfully compressed the variation of one of the variables. A suggestive pattern emerges nonetheless: although the OLS beta is essentially nil, the instrumental-variables estimators recover a beta twelve times larger in absolute value

($\hat{\beta}_{IVX} = \hat{\beta}_{RA-IVX} = -0,024$), and we recover some significance with a more negative t -statistic. In this case, the positive Stambaugh bias visibly attenuates OLS toward zero. The signal remains too weak to clear conventional significance thresholds, but the direction of the bias correction is exactly what theory predicts. We might still ask, however, why the advantage documented in the Monte Carlo simulation does not show up here. As Table 4 shows, the parameter $\hat{\gamma}$ is 0.025 in this subsample, essentially nil. The augmentation corrects by subtracting $\hat{\gamma}\hat{v}_t$ from the regression, meaning there is little left to correct. This happens because inflation during the Great Moderation is very low-volatility relative to unemployment, which pushes γ toward zero even though some correlation between the series remains.

Finally, the Bernanke/Yellen/Powell subsample is where the methodological contribution becomes visible. We can see that both OLS and IVX fail to reach significance, with $t_{OLS} = -1,305$ and a somewhat better t_{IVX} of $-1,439$. RA-IVX, however, appears to recover a signal that neither OLS nor IVX detects: we reach significance at the 10 % level with a t_{RA-IVX} of $-1,969$. With $\widehat{\text{corr}} = -0,103$, endogeneity is not negligible, and the reported beta is negative, exactly the setting in which the simulations gave RA-IVX its largest advantage.

Here only one of the two channels explained in Section 2 is at work. Correcting the positive bias gives us a more negative $\hat{\beta}$ ($-0,017$ versus $-0,012$ for IVX), which pushes our t -statistic toward the rejection region. The second channel, however, does not materialize: the variance reduction stemming from $\sigma_\varepsilon^2 < \sigma_u^2$. RA-IVX's standard error is not smaller than IVX's in this subsample, because the gain is offset by the efficiency cost of estimating $\hat{v}_t$.

4. Conclusion

This thesis has examined the econometric challenges that arise in predictive regressions with highly persistent regressors, working through the literature from the Stambaugh bias, through local-to-unity asymptotics and bias-correction methods, to the IVX framework and the residual-augmented extension of Demetrescu and Rodrigues (2022). Building on this methodological foundation, we examined the finite-sample properties of OLS, IVX, and RA-IVX through Monte Carlo simulation, and applied these methods to the question of Phillips-curve predictability.

The simulations suggest that under positive alternatives, two opposing effects of using

RA-IVX can offset one another, so that the advantage is partly attenuated. Under negative alternatives, these two effects do not offset each other, so we would expect a larger advantage. Our empirical application with the Bernanke/Yellen/Powell subsample is consistent with that documented advantage materializing in real data, recovering significance relative to both IVX and OLS.

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5. Tables and Figures

Cuadro 1: Empirical size of the tests under $H_0 : \beta = 0$, for different degrees of persistence ρ and sample sizes n .

n	ρ	OLS	IVX	RA-IVX
50	0.70	0.066	0.072	0.053
	0.90	0.100	0.073	0.042
	0.95	0.160	0.080	0.042
	0.99	0.219	0.066	0.034
100	0.70	0.058	0.065	0.059
	0.90	0.072	0.058	0.041
	0.95	0.106	0.066	0.038
	0.99	0.199	0.059	0.035
200	0.70	0.056	0.061	0.056
	0.90	0.065	0.066	0.052
	0.95	0.078	0.064	0.038
	0.99	0.158	0.058	0.028
500	0.70	0.051	0.050	0.046
	0.90	0.069	0.064	0.062
	0.95	0.052	0.046	0.038
	0.99	0.102	0.059	0.031

Cuadro 2: Monte Carlo results: mean bias and distribution of the t -statistic ($n = 100$, $\rho = 0,99$, $\rho_{uv} = -0,9$, $N_{sim} = 5000$).

β_{true}	Estimator	$\bar{\hat{\beta}}$	Mean bias	$\bar{t}$	$\overline{SE}$
-0,075	IVX	-0,0328	0,0422	-1,1551	0,0471
	RA-IVX	-0,0576	0,0174	-1,9073	0,0371
0	IVX	0,0437	0,0437	0,7852	0,0476
	RA-IVX	0,0171	0,0171	0,3767	0,0379
0,075	IVX	0,1177	0,0427	2,6761	0,0473
	RA-IVX	0,0903	0,0153	2,5550	0,0383

Cuadro 3: Empirical power of the tests for $n = 50$, full β grid. We show the smallest sample size with the complete β grid since it is where the differences between estimators are most pronounced; for larger n , all three estimators converge quickly to the same V-shaped profile (see Figure 2)

β	OLS	IVX	RA-IVX
-0.150	0.312	0.250	0.458
-0.125	0.223	0.171	0.332
-0.100	0.141	0.117	0.217
-0.075	0.100	0.063	0.135
-0.050	0.096	0.050	0.066
-0.025	0.124	0.042	0.030
0.000	0.219	0.066	0.034
0.025	0.361	0.116	0.070
0.050	0.608	0.203	0.130
0.075	0.825	0.375	0.265
0.100	0.919	0.575	0.423
0.125	0.980	0.711	0.579
0.150	0.993	0.831	0.711

Cuadro 4: Persistence and Endogeneity Diagnostics

Sample	T	Inflation		Unemployment		Endogeneity	
		AR(1)	ADF p	AR(1)	ADF p	$\widehat{\text{corr}}(u, v)$	$\hat{\gamma}$
<i>Panel A: Euro Area — Monthly</i>							
Full sample	299	0.413	<0.05	1.003	0.220	-0.022	0.062
Pre-2008	93	0.136	<0.01	1.025	0.274	-0.064	0.539
Post-2008	202	0.453	<0.05	1.002	0.262	-0.032	0.013
<i>Panel B: United States — Monthly</i>							
Full sample	795	0.621	<0.01	0.968	<0.05	-0.041	-0.072
Pre-Volcker	241	0.648	0.248	0.989	0.181	0.086	0.185
Volcker-Greenspan	313	0.605	<0.01	0.995	0.183	0.027	0.077
Great Moderation	283	0.320	<0.01	0.987	0.175	-0.039	0.025
Bernanke/Yellen/Powell	230	0.540	<0.01	0.942	0.189	-0.103	-0.108

Notes: AR(1) estimated by OLS: $x_t = a + \rho x_{t-1} + u_t$. ADF test with trend (drift); lag length = $\lfloor n^{1/3} \rfloor$; p -values are based on the critical values of MacKinnon (1994). $\widehat{\text{corr}}(u, v)$ denotes the estimated correlation between the predictive regression's disturbance and the innovation of the unemployment autoregressive process. $\hat{\gamma}$ is the OLS coefficient from projecting $\tilde{\pi}_{t+1}$ onto $\hat{v}_t$ in the RA-IVX augmentation stage.

Cuadro 5: Summary of Predictive Regression Results

Sample	N	Estimated coefficients (SE)			t -statistics			$\widehat{\text{corr}}(u, v)$
		$\hat{\beta}_{\text{OLS}}$	$\hat{\beta}_{\text{IVX}}$	$\hat{\beta}_{\text{RA-IVX}}$	t_{OLS}	t_{IVX}	$t_{\text{RA-IVX}}$	
<i>Panel A: Euro Area — Monthly</i>								
Full sample	299	−0,0434** (0,0086)	−0,0419** (0,0113)	−0,0419** (0,0113)	−5,049**	−3,721**	−3,712**	−0,022
Pre-2008	93	−0,0385 (0,0283)	−0,0362 (0,0386)	−0,0365 (0,0397)	−1,361	−0,938	−0,918	−0,064
Post-2008	202	−0,0425** (0,0100)	−0,0423** (0,0092)	−0,0423** (0,0092)	−4,234**	−4,584**	−4,583**	−0,032
<i>Panel B: United States — Monthly</i>								
Full sample	795	0,0032 (0,0066)	0,0083 (0,0059)	0,0063 (0,0060)	0,494	1,407	1,051	−0,041
Volcker-Greenspan	313	0,0262* (0,0103)	0,0298* (0,0118)	0,0304** (0,0118)	2,537*	2,528*	2,579**	0,027
Great Moderation	283	−0,0020 (0,0127)	−0,0241 (0,0319)	−0,0238 (0,0319)	−0,158	−0,756	−0,747	−0,039
Bernanke/Yellen/Powell	230	−0,0124 (0,0095)	−0,0119 (0,0083)	−0,0167* (0,0085)	−1,305	−1,439	−1,969*	−0,103

Notes: The table reports the estimated coefficients of the OLS, IVX, and RA-IVX predictive regressions of inflation on unemployment, with the standard error in parentheses immediately below each coefficient, the corresponding t -statistic, and the sample size N . $\widehat{\text{corr}}(u, v)$ denotes the estimated correlation between the predictive regression's disturbance and the innovation of the unemployment autoregressive process (common to all three estimators within each subsample). Significance levels are indicated by *, **, and *** (10 %, 5 %, and 1 %, respectively).

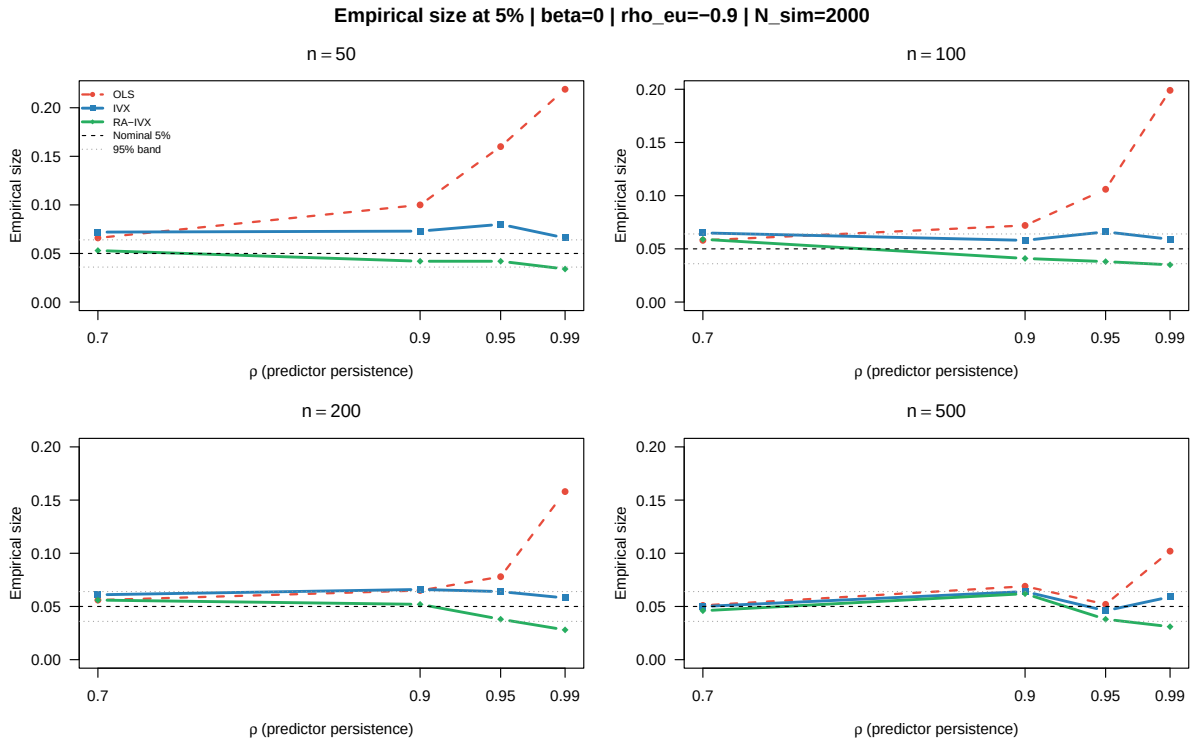


Figure 1: Figure 1. Empirical size at the 5% level. $\beta = 0$, $\rho_{uv} = -0.9$, $N_{sim} = 2,000$.

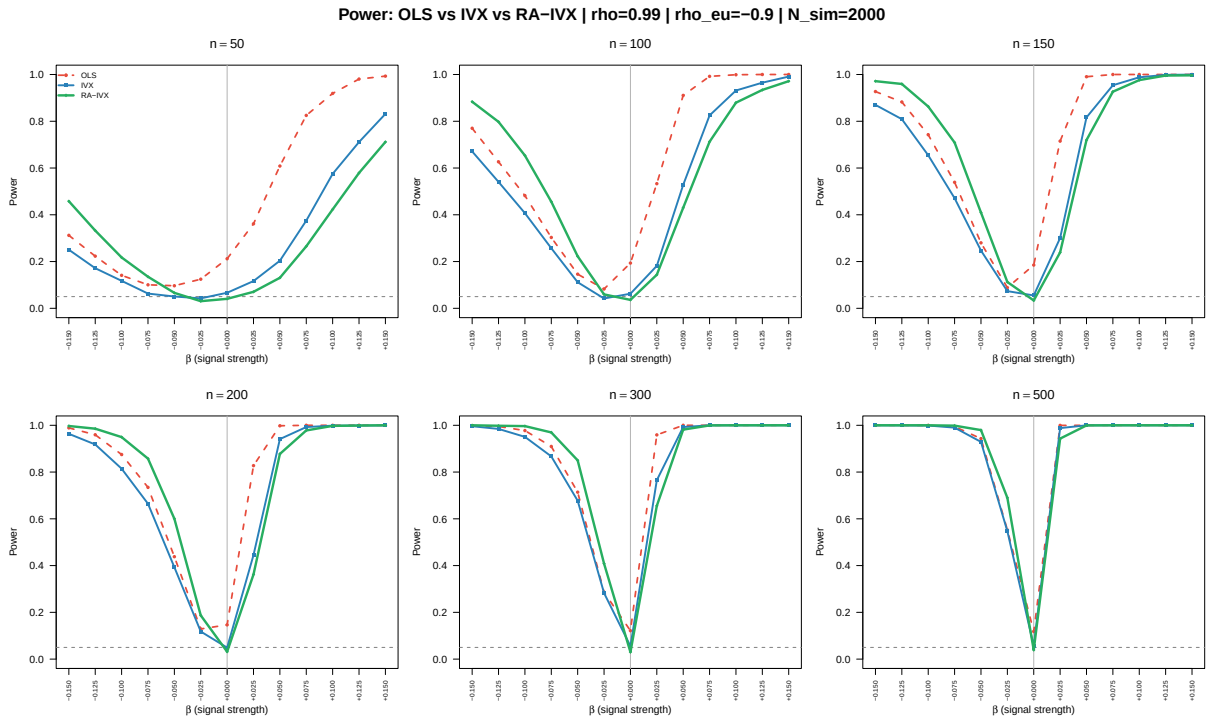


Figure 2: Figure 2. Empirical power at the 5% level. $\rho = 0.99$, $\rho_{uv} = -0.9$, $N_{sim} = 2,000$.

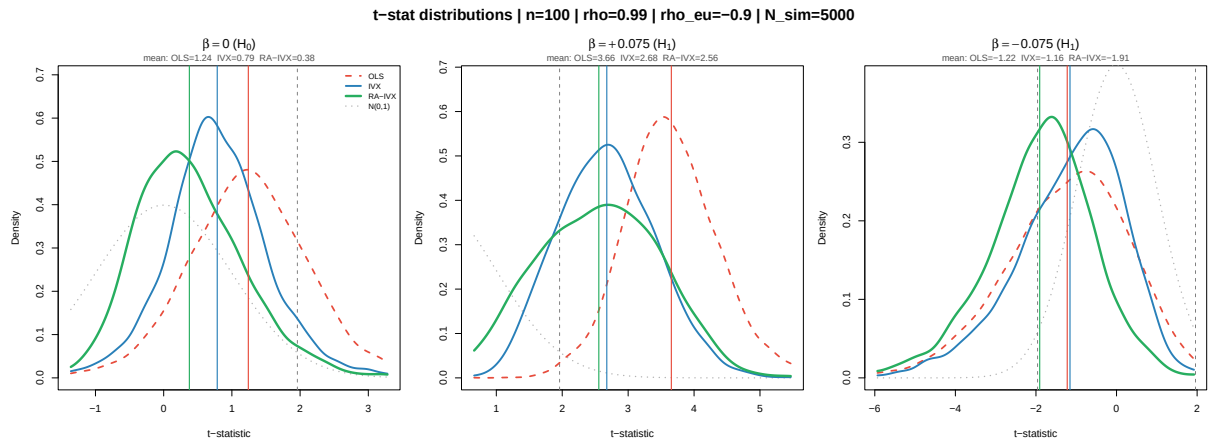


Figure 3: Finite-sample t -statistic distributions. $\rho = 0,99$, $N_{\text{sim}} = 5,000$.

A. The Stambaugh–Kendall Bias Decomposition

We start from the OLS estimator of β in the predictive regression. Subtracting the population value, the estimation error can be written purely in terms of the return innovation:

$$\hat{\beta} - \beta = \frac{\sum x_t u_{t+1}}{\sum x_t^2} \quad (7)$$

The bias arises because u_{t+1} and the predictor share information. To see this, we split u_{t+1} into the part tied to the dynamics of x and the part that is not. Projecting u_{t+1} onto v_{t+1} gives $u_{t+1} = \gamma v_{t+1} + \varepsilon_{t+1}$, with $\gamma = \sigma_{uv}/\sigma_v^2$ and $\varepsilon_{t+1} \perp v_{t+1}$ by construction. Substituting, we obtain Stambaugh’s decomposition:

$$\hat{\beta} - \beta = \gamma \cdot \frac{\sum x_t v_{t+1}}{\sum x_t^2} + \frac{\sum x_t \varepsilon_{t+1}}{\sum x_t^2} \quad (8)$$

The two terms behave differently. ε_{t+1} is, by construction, the part of u_{t+1} orthogonal to v_{t+1} . This means it bears no relation to the predictor’s dynamics. Since $x_{t+1} = \rho x_t + v_{t+1}$, the only thing that moves x forward is a v shock, so ε_{t+1} leaves no trace in any x_s and therefore leaves no trace in the denominator $\sum x_s^2$. This means that, writing A_ε for its numerator and B for the denominator, $\text{Cov}(A_\varepsilon, B) = 0$. The first term will not contribute to the bias. The term in v is different. v_{t+1} is the predictor’s own shock, so it filters into $x_{t+1}, x_{t+2}, \dots$ and never leaves the denominator $\sum x_s^2$ through the future values of x . The numerator and denominator share the same shock, so they co-move. In fact, the ratio $\sum x_t v_{t+1} / \sum x_t^2$ is exactly the OLS estimation error of ρ in the AR(1):

$$\hat{\rho} - \rho = \frac{\sum x_t v_{t+1}}{\sum x_t^2} \quad (9)$$

The bias in $\hat{\beta}$ therefore follows directly from the bias in $\hat{\rho}$, scaled by γ ; Kendall (1954) provides the closed-form analytical expression $\mathbb{E}[\hat{\rho} - \rho] \approx -(1 + 3\rho)/T$ under near-unit-root behavior.

B. Phillips Curve Estimation Strategy

OLS is estimated by ordinary least squares on the demeaned series $\tilde{\pi}_{t+1}$ and $\tilde{x}_t$, with conventional homoskedastic standard errors, and is included as a biased benchmark. The IVX instrument is built recursively as

$$z_t = \rho_z z_{t-1} + \Delta x_t, \quad z_0 = 0, \quad \rho_z = 1 - T^{-0.95}, \quad (10)$$

following the calibration of Kostakis, Magdalinos and Stamatogiannis (2015). The point estimator and standard errors are:

$$\hat{\beta}_{\text{IVX}} = \frac{\sum_t z_{t-1} \tilde{\pi}_{t+1}}{\sum_t z_{t-1} \tilde{x}_t}, \quad \widehat{\text{se}}_{\text{IVX}} = \frac{\sqrt{\hat{\sigma}_u^2 \cdot \sum_t z_{t-1}^2}}{|\sum_t z_{t-1} \tilde{x}_t|}. \quad (11)$$

where $\hat{\sigma}_u^2 = n^{-1} \sum_t \hat{u}_{t+1}^2$ is computed from the OLS residuals of $\tilde{\pi}_{t+1}$ on $\tilde{x}_t$.

RA-IVX. Proceeds in two steps prior to applying IVX.

Step 1 — residual augmentation. An $\text{AR}(p)$ is fitted to $\tilde{x}_t$ by OLS:

$$\tilde{x}_t = \sum_{j=1}^p \hat{\rho}_j \tilde{x}_{t-j} + \hat{v}_t, \quad (12)$$

with p selected by AIC over $p \in \{0, \dots, p_{\max}\}$, $p_{\max} = \lfloor 4(T/100)^{1/4} \rfloor$. We project $\tilde{\pi}_{t+1}$ onto $\hat{v}_t$ and orthogonalize:

$$\tilde{\pi}_{t+1} = \tilde{\pi}_{t+1} - \hat{\gamma} \hat{v}_t, \quad \hat{\gamma} = \frac{\sum_t \hat{v}_t \tilde{\pi}_{t+1}}{\sum_t \hat{v}_t^2}. \quad (13)$$

Step 2. IVX is applied to $\tilde{\pi}_{t+1}$ on $\tilde{x}_t$ using the same instrument z_t , obtaining $\tilde{\beta}_{\text{IVX}}$. The variance estimator is robust to heteroskedasticity:

$$\widehat{\text{Var}}(\tilde{\beta}_{\text{IVX}}) = \frac{\sum_t z_{t-1}^2 \tilde{\varepsilon}_{t+1}^2 + \hat{\gamma}^2 \hat{Q}}{(\sum_t z_{t-1} \tilde{x}_t)^2}, \quad (14)$$

where $\tilde{\varepsilon}_{t+1}$ are the residuals from the joint OLS projection of $\tilde{\pi}_{t+1}$ on $(\tilde{x}_t, \hat{v}_t)$, and $\hat{Q}$ is the sandwich-type correction for the estimation uncertainty in $\hat{\gamma}$.